

Identifying Persuadable Voters with and without Social Media

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Business Problem

This project develops and compares two uplift models to identify the top 10% of persuadable voters within a population during election campaigns. The first model used only anonymized marketing campaign data, and the second model built on this data with tweets from the six months leading up to the 2024 presidential election. The comparison is meant to inform users about the richness of social media data and its application in uplift models designed to maximize cost-efficiency and vote effectiveness.

Background/History

Data Science has been a key tool in elections since Obama's first campaign. Uplift models are commonly used in business to identify customers who only purchase when contacted. Using this model to target persuadable voters aims to increase cost savings by conserving precious campaign dollars while still maximizing the candidate's vote share. Uplift models can identify four voter segments: persuadables (only vote if contacted), sure things (vote regardless of contact), lost causes (won't vote regardless of contact), and do-not-disturbs (won't vote if contacted). By using the model to identify the top 10% of persuadables, the maximum impact can be made with the fewest number of contacts and the least dollars spent.

Data Explanation

The datasets to be used for this project are the *Uplift Modeling, Marketing Campaign Data* dataset, and the *2024 US Presidential Elections Twitter Data* dataset, both sourced from Kaggle. It's important to note that voter data is generally protected for individuals' privacy, and even anonymized voter datasets are difficult to obtain. For this reason, the first dataset will be used to simulate voter data.

It was created by the Criteo AI Lab and was collected in a randomized control trial for the digital advertising industry. It "consists of 13M rows, each one representing a user with 12 features, a treatment indicator, and 2 binary labels (visits and conversions)" (Mobius, 2020). The 12 features are represented as

dense float values, and the feature names have been replaced with feature numbers. This, along with randomizing both the sample and the values, is an effort to protect users' privacy while maintaining predictive power. It's reasonable to assume the features likely contain demographic information similar to that in an anonymized voter dataset, along with user activity data. The content section on the Kaggle page noted that positive labels indicate "the user visited/converted on the advertiser website during the test period (2 weeks)" (Mobius, 2020).

As for the Twitter dataset, the author provided several options for the user. All of the datasets included tweets from the six months leading up to the 2024 election, obtained via web scraping using the following phrases: 2024 election, 2024 US election, Biden, Kamala Harris, Trump, Democrat, Republican, Vivek Ramaswamy, JD Vance, Mike Pence, Nikki Haley, and RFK Jr (Row, 2025). There were three different raw sets: Democrat, Republican, and both combined. There were also two preprocessed sets. Both have implemented emoji replacement, removal of HTML and URL tags, chat word replacement, expansion of contractions, and removal of special characters. One of the preprocessed files includes all tweets translated into English. The translated, preprocessed set included the following variables: Tweet ID, Username, Text, Date, #Retweets, Likes, #Weight, and has 26,353 entries (Row, 2025).

Methods

The data in both datasets required minimal preprocessing, as this step had already been taken care of by the uploaders at Kaggle. The first step in this analysis was to create a matched sample from the marketing campaign dataset, which was much larger than the Twitter dataset. Care was taken to preserve the original treatment and conversion ratios in the marketing campaign data by creating a stratified sample. This was accomplished by creating a column indicating each row's stratum (e.g., 1-0 indicates that the row was treated and did not convert), and then only pulling rows to create a dataframe with the same

composition. A variable report was then run for both datasets to flag any potential issues (such as duplicates, skew, or imbalance).

The variable report for the marketing campaign data showed some evidence of multicollinearity among the dense float features, a negligible number of duplicates, and an imbalance in the conversion, treatment, and visit variables. All three variables heavily favor the 0 value, meaning that it's rare to convert (2.9% do), visit (27.3% do), or be exposed (19.5% are). These imbalances were preserved from the original dataset, and are consistent with real-world uplift modeling scenarios, where outcomes and behavioral events are typically sparse. There were no alerts for heavy skew within the dataset, indicating that the float predictors are reasonably symmetric and don't require transformation. There were 7 duplicates in the 26,353-row dataset, representing less than 0.1%. The duplicates were preserved rather than dropped, to maintain all 26,353 rows for analysis. The variable report for the Twitter data didn't reveal any alerts for the text column, which was the only column used in this analysis.

Feature selection was conducted on the Twitter text data using SentenceTransformer, a miniLM distilled from BERT. SentenceTransformer performs better out of the box for sentence embeddings, as it has been optimized for this task while still sharing the same basic architecture as BERT. After creating the embeddings, PCA was used to compress the high-dimensional results to their most important features, creating a more digestible dataframe for the uplift modeling.

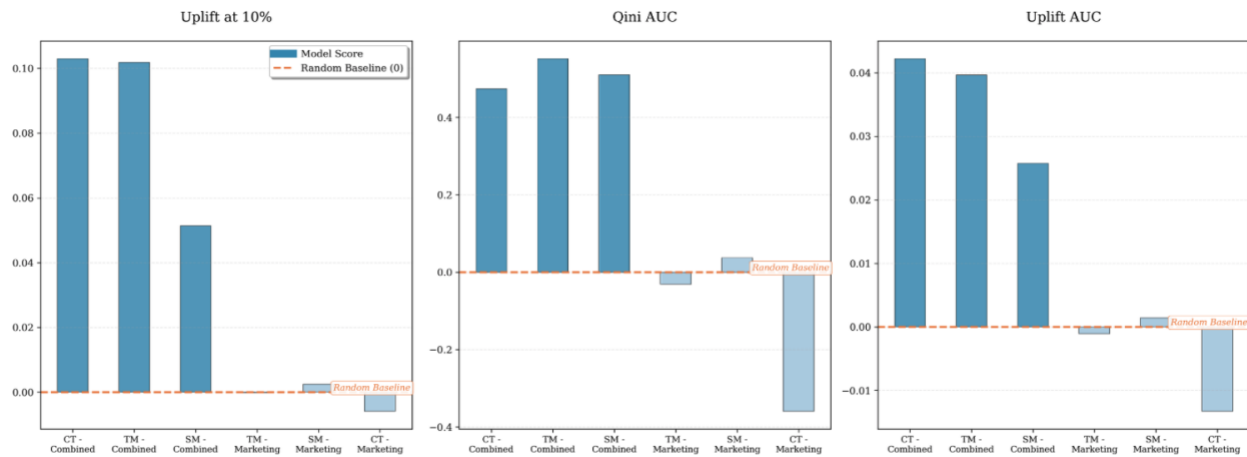
After merging the dataframes, the datasets were stratified by treatment and conversion, yielding an 80/20 split for model training and evaluation. Including the treatment was necessary to properly evaluate the models. Then, GridSearchCV was used to select the best model and hyperparameters for both datasets, choosing between scikit-learn's Solo Model, Class Transformation, and Two Model, all built upon a Logistic Regression model base. GridSearchCV was used in this case for reproducibility, even though RandomizedSearchCV is known to often yield a better-tuned model.

Analysis

The evaluation metrics clearly showed that adding the Twitter data to the marketing campaign data enriched the signal. The evaluation metrics were sorted by their uplift score at 10%, as determined by the selection metric (see Appendix A). The marketing-only models had the three lowest scores across all metrics (see Figure 1).

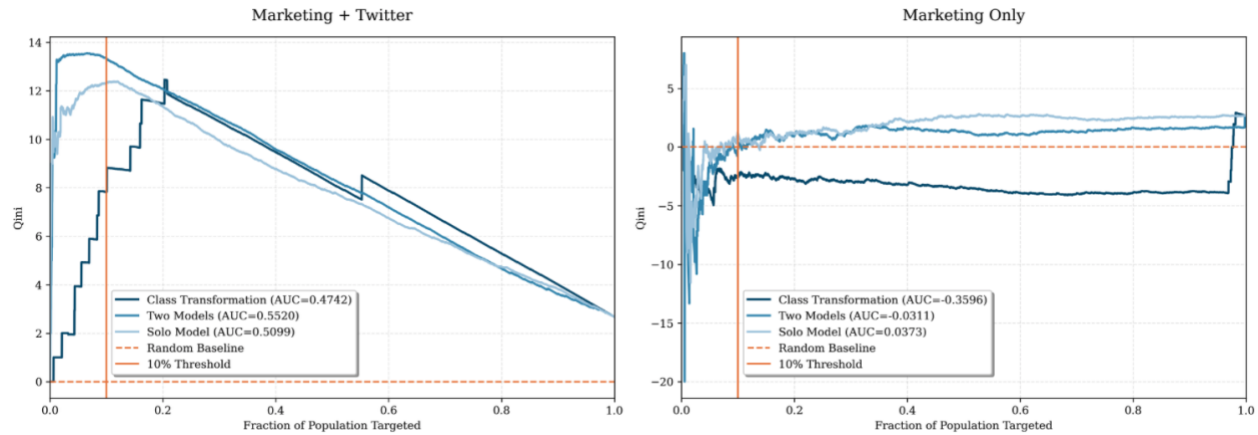
Figure 1

Uplift Model Evaluation Metrics



Note. Models trained on the combined marketing campaign and Twitter datasets outperformed marketing-only models across all metrics.

The Class Transformation model with the combined datasets had the highest uplift score (0.1030) at the 10% evaluation metric. This can be interpreted to mean that the top 10% of persuadables identified by the model had a 10.30 percentage-point higher conversion rate due to treatment. Although the combined dataset Two Model has the highest Qini AUC score, meaning it's better at ranking persuadables across the entire population, the combined dataset Class Transformation still dominated the Uplift AUC score. Qini curves were plotted for both datasets and all models (see Figure 2).

Figure 2*Qini Curves by Dataset*

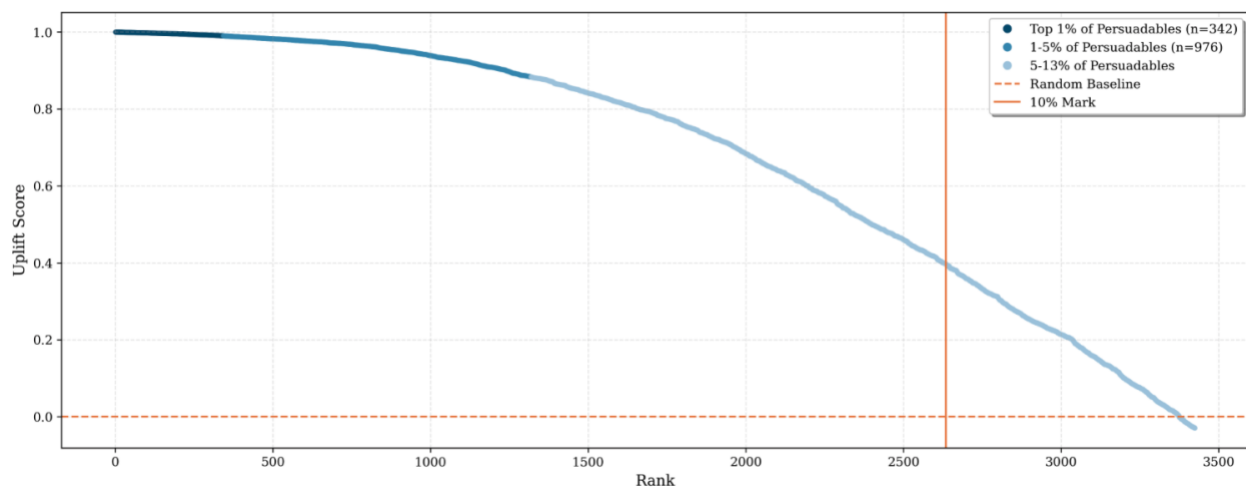
Note. The Qini curves showed a stark contrast between the combined datasets and the stand-alone marketing dataset.

All three models for the combined datasets performed better than the random baseline in the first 10% on the Qini curves, and then tapered off in the remaining 90%. This means the model is doing a good job of identifying the voters most likely to respond to treatment, but its targeting power drops the further you get from the top 10%. The marketing-only models either barely exceeded the random baseline or performed worse than it. Because the purpose of this project was to identify the top 10% of persuadables, the combined Class Transformation model will be used to predict them, given its higher uplift score at 10%, even though it had the weakest Qini curve performance among the three combined models. The class transformation model also scored highest in the uplift AUC metric, with the highest area beneath the uplift curve. The holdout scores and CV average scores are close across all models, suggesting minimal overfitting. The training and CV scores for the best model also have a minimal gap, further supporting this idea (see Appendix B).

After generating the persuadable rankings, the summary of the top 10% of raw predicted uplift scores shows strong uplift potential (mean: 0.818) compared to random selection. With a minimum of 0.396268, targeting could be expanded slightly based on resource capacity and ROI calculations (see Appendix C). You can see that the uplift scores remain well above the random baseline but taper off by ~13%. (see Figure 3).

Figure 3

Uplift Score Distribution – Top 13% Persuadables



Note. The raw predicted uplift scores remain well above the random baseline threshold throughout the first 10%, and start to drop below the line around 13%. The top 1% has nearly perfect scores, and the top 1-5% remain above 0.80.

Conclusion

The project demonstrated that using the embedded and compressed 2024 election Twitter data enriched the marketing campaign data sample in a meaningful way. Uplift models are unique in that you often expect minimal signal in noisy data, yet they can still deliver incremental results that create data-driven change when used wisely. The results from combining the marketing campaign and Twitter data

suggested that expanding reach to nearly the top 13% of the population is possible, provided that the uplift score $\times$ average conversion value exceeds the campaign cost per person.

Assumptions

Since pre-processed data were used in this analysis, the project relied on the assumption that this data was valid and processed correctly. Even though the marketing campaign dataset was stratified, the sample assumes that it preserves the signal from the much larger original dataset (which contained 13 million samples). Because of this dilution of the original marketing dataset, it's possible that a portion of the signal was overloaded by the embedded and condensed Twitter features, exaggerating the Twitter data's effect on the model.

Limitations/Challenges

It's important to note that the data used in this project can only be interpreted as a simulation of the mechanical process, due to the previously mentioned sampling assumption and the combination of the two datasets. The datasets were combined so that tweets were arbitrarily assigned to the consumer data. Even though the results suggested a deployable model, the importance of its features cannot be accurately assessed, because they don't come from the same consumer or voter.

Future Use/Additional Applications

Based on this mechanical simulation, it's reasonable to predict that the same pipeline applied to real-world voter and Twitter data could yield similar results. It is important to note that examining each population's rank score distribution is necessary to optimize cost-effectiveness and vote-efficiency within that population. The same pipeline may also be useful for targeted marketing campaigns that aren't election-specific, such as donation campaigns or targeting populations for life-saving health screenings that wouldn't seek diagnosis without being prompted.

Recommendations

Using the same pipeline on real-world data can provide an opportunity that didn't exist within this project: examining feature importance. The marketing data didn't include feature definitions to protect consumer privacy, but when working with real-world data, answers to what makes a person persuadable lie in these feature importances. I would recommend leaning into this idea in future implementations. The low signal in the marketing data could also have been due to using a linear algorithm on data that showed signs of multicollinearity in the data exploration phase, so I'd be interested in running the project pipeline again with tree-based models.

Implementation Plan

The uplift models were intentionally built on logistic regression because they are simple to run and interpretable. To be implemented in a real-world scenario, a data preprocessing pipeline would need to be added for reproducibility and scalability. With uplift models, you have to collect data on the population in question before even running the model, so access to private voter information and social media scraping capabilities would be necessary.

Ethical Assessment

The uplift dataset has already been anonymized, so privacy wasn't a concern. The Twitter dataset includes a username feature that wasn't used in this analysis, preserving individual users' anonymity. In the larger picture, the question of "Is using a consumer's data to identify if they're persuadable ethical?" I think a case can be made for both sides. This is now a common tactic in elections, meaning that all sides have access to and use these tools. Equal access levels the playing field, but in a staunchly divided country where swing votes decide elections, I worry that zeroing in on a small population isn't representative of the country as a whole. As previously mentioned in the statement on feature importance, I think it would

be meaningful to conduct further research into what makes the persuadable *persuadable* and how we can harness that knowledge to foster greater understanding and open-mindedness across party lines.

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Appendix A

Evaluation Metrics Table

Model	Qini AUC	Uplift AUC
Marketing + Twitter - Class Transformation	0.4742	0.0422
Marketing + Twitter - Two Models	0.5520	0.0396
Marketing + Twitter - Solo Model	0.5099	0.0257
Marketing Only - Solo Model	0.0373	0.0014
Marketing Only - Two Models	-0.0311	-0.0011
Marketing Only - Class Transformation	-0.3596	-0.0133
Uplift at 10%		
0.1030		
0.1018		
0.0514		
0.0025		
-0.0001		
-0.0059		

Note. The evaluation metrics were sorted by their uplift score at 10%, as determined by the selection metric. Models trained on the combined marketing campaign and Twitter datasets outperformed marketing-only models across all metrics.

Appendix B

RandomizedSearchCV Results – Comparing CV Average and Holdout Score

```
Fitting 5 folds for each of 7 candidates, totalling 35 fits
Marketing Only | Solo Model: CV Average = 0.0051, Holdout Score = 0.0025, Best Params = {'estimator__C': 0.1, 'estimator__penalty': 'l2', 'estimator__solver': 'liblinear'}
Fitting 5 folds for each of 7 candidates, totalling 35 fits
Marketing Only | Class Transformation: CV Average = -0.0002, Holdout Score = -0.0059, Best Params = {'estimator__C': 0.1, 'estimator__penalty': 'l2', 'estimator__solver': 'liblinear'}
Fitting 5 folds for each of 49 candidates, totalling 245 fits
Marketing Only | Two Models: CV Average = 0.0097, Holdout Score = -0.0001, Best Params = {'estimator__C': 0.5, 'estimator__penalty': 'l2', 'estimator__solver': 'liblinear', 'estimator__trmmt__C': 10.0, 'estimator__trmmt__penalty': 'l2', 'estimator__trmmt__solver': 'liblinear'}
Fitting 5 folds for each of 7 candidates, totalling 35 fits
Marketing + Twitter | Solo Model: CV Average = 0.0575, Holdout Score = 0.0514, Best Params = {'estimator__C': 10.0, 'estimator__penalty': 'l2', 'estimator__solver': 'liblinear'}
Fitting 5 folds for each of 7 candidates, totalling 35 fits
Marketing + Twitter | Class Transformation: CV Average = 0.1299, Holdout Score = 0.1030, Best Params = {'estimator__C': 2.0, 'estimator__penalty': 'l2', 'estimator__solver': 'liblinear'}
Fitting 5 folds for each of 49 candidates, totalling 245 fits
Marketing + Twitter | Two Models: CV Average = 0.0850, Holdout Score = 0.1018, Best Params = {'estimator__C': 10.0, 'estimator__penalty': 'l2', 'estimator__solver': 'liblinear', 'estimator__trmmt__C': 9.36, 'estimator__trmmt__penalty': 'l2', 'estimator__trmmt__solver': 'liblinear'}
```

Train: 0.1021 | CV: 0.0850 | Gap: 0.0171

Note. The holdout scores and CV average scores are close across all models, suggesting minimal overfitting. The training and CV scores for the best model also have a minimal gap, further supporting this idea.

Appendix C

Uplift Score Summary for Top 10% Persuadables

```
Best model loaded: ClassTransformation(estimator=LogisticRegression(C=2.0, random_state=0,
                                                                    solver='liblinear'))
Total population: 26353
Top 10% persuadables identified: 2635
```

```
Uplift score summary for top 10%:
count    2635.000000
mean      0.817996
std       0.179298
min       0.396268
25%       0.695109
50%       0.883201
75%       0.973875
max       0.999859
```

Note. The summary of the top 10% raw predicted scores shows strong uplift potential (mean: 0.818) compared to random selection. With a minimum of 0.396268, targeting could be expanded up to the top 13% based on resource capacity and ROI calculations.